# ORTHOGONALITY OF THE CONE ENTRY VERSUS ISOMORPHISM OF HOM OUT OF THE SUBCATEGORY

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*Editorial note.* Counted span is the author's own proof environment, source lines 11915-11932, following the statement at 11899-11913.

*Editorial note.* No omissions: both exact sequences are displayed.

*Allowed standard prerequisites (editorial).*

- Representable functor is homological. lemma-representable-homological (same chapter, cited at ~11917). The long exact Hom sequences used in both directions.

*Source reference.* Original equation 0148: sources/stacks-derived/pinned-original-source.tex, source lines 499–499. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 0149: sources/stacks-derived/pinned-original-source.tex, source lines 603–603. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *Let $\mathcal{D}$ be a triangulated category. Let $\mathcal{A} \subset \mathcal{D}$ be a full subcategory invariant under all shifts. Consider a distinguished triangle*

$$X \to Y \to Z \to X[1]$$

*of $\mathcal{D}$. The following are equivalent*

(1) *$Z$ is in $\mathcal{A}^\perp$, and*
(2) *$\mathrm{Hom}(A, X) = \mathrm{Hom}(A, Y)$ for all $A \in \mathrm{Ob}(\mathcal{A})$.*

*Proof.* By Lemma 0149 the functor $\mathrm{Hom}(A, -)$ is homological and hence we get a long exact sequence as in (0148). Assume (1) and let $A \in \mathrm{Ob}(\mathcal{A})$. Then we consider the exact sequence

$$\mathrm{Hom}(A[1], Z) \to \mathrm{Hom}(A, X) \to \mathrm{Hom}(A, Y) \to \mathrm{Hom}(A, Z)$$

Since $A[1] \in \mathrm{Ob}(\mathcal{A})$ we see that the first and last groups are zero. Thus we get (2). Assume (2) and let $A \in \mathrm{Ob}(\mathcal{A})$. Then we consider the exact sequence

$$\mathrm{Hom}(A, X) \to \mathrm{Hom}(A, Y) \to \mathrm{Hom}(A, Z) \to \mathrm{Hom}(A[-1], X) \to \mathrm{Hom}(A[-1], Y)$$

and we conclude that $\mathrm{Hom}(A, Z) = 0$ as desired. □

*Date*: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.